Homework

DM5 - Splitting an area - 55 min, after S79.

Consider the function f defined on $[0;\frac{\pi}{2}]$ by $f(x)=\sin x\cos x$. Its graph is drawn in a orthonormal coordinate system.

1. Calculate $f'(x)$ and determine the maximum of f over this interval.

2. Solve on $[0;\frac{\pi}{2}]$ the inequality $f(x)\geq\frac{1}{4}$.

3. Calculate $\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x$, then the average value of f on $[0;\frac{\pi}{2}]$.

4. For $a\in[0;\frac{\pi}{2}]$, set $A(a)=\int_0^a f(x)\,\mathrm{d}x$. Determine a so that the region under the curve between 0 and a has an area equal to half the total area under the curve between 0 and $\frac{\pi}{2}$.